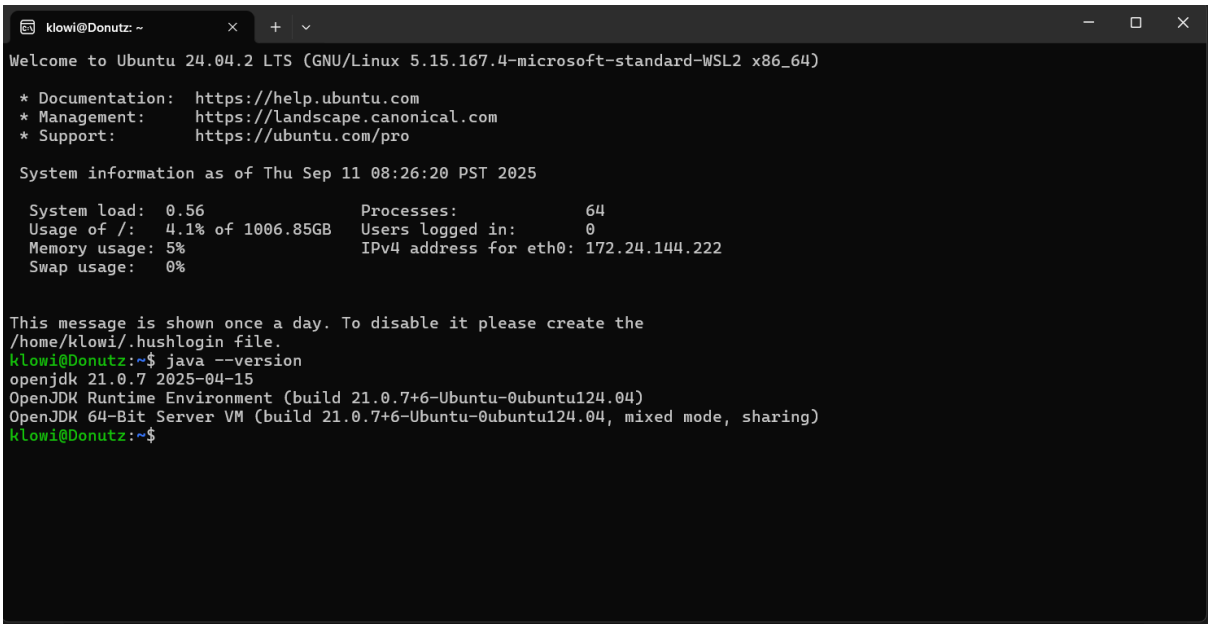
CO1 FA2 Kafka

Members: Dela Cruz, Cloie Sofia & Respecio, Luis Arnold N.

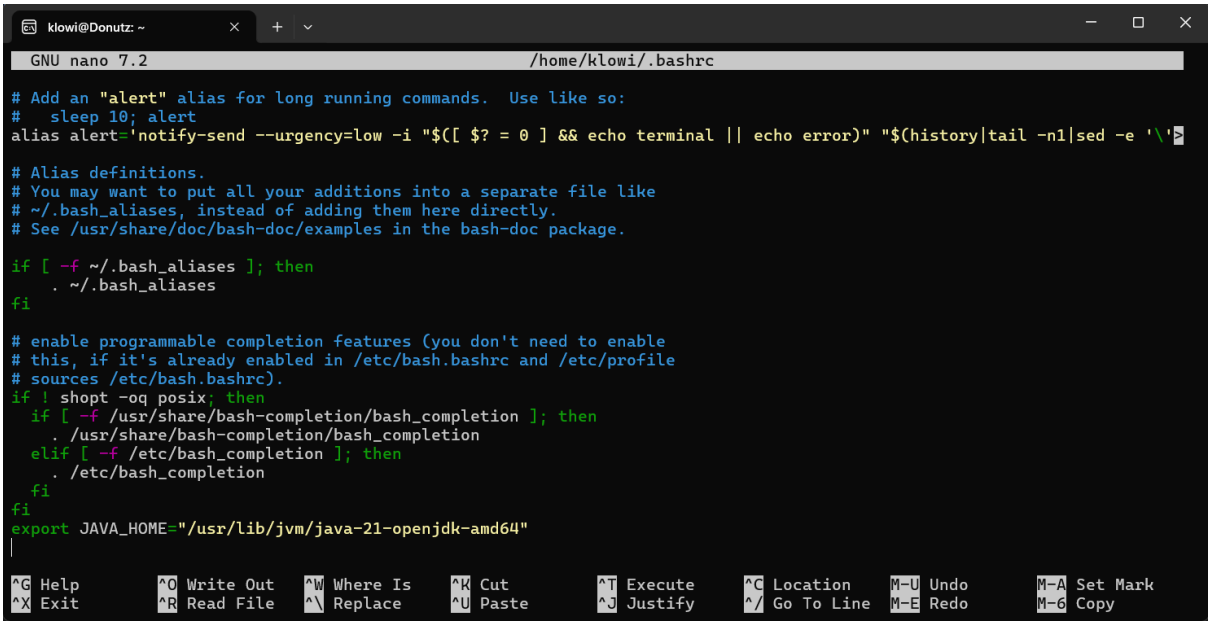
Java Installation & Setup



Java will be needed for setting up Apache Kafka. If Java is not yet installed, install it. In this case, Java is already updated and available. In here, Java was checked for its availability and version. It can be seen that OpenJDK is available with version 21.



As said, the environment variable was set through the nano text editor in WSL. This opened the .bashrc file and displayed a whole body of text, where the environment is configured for a customized Bash shell session every time (Verma et al., 2025).

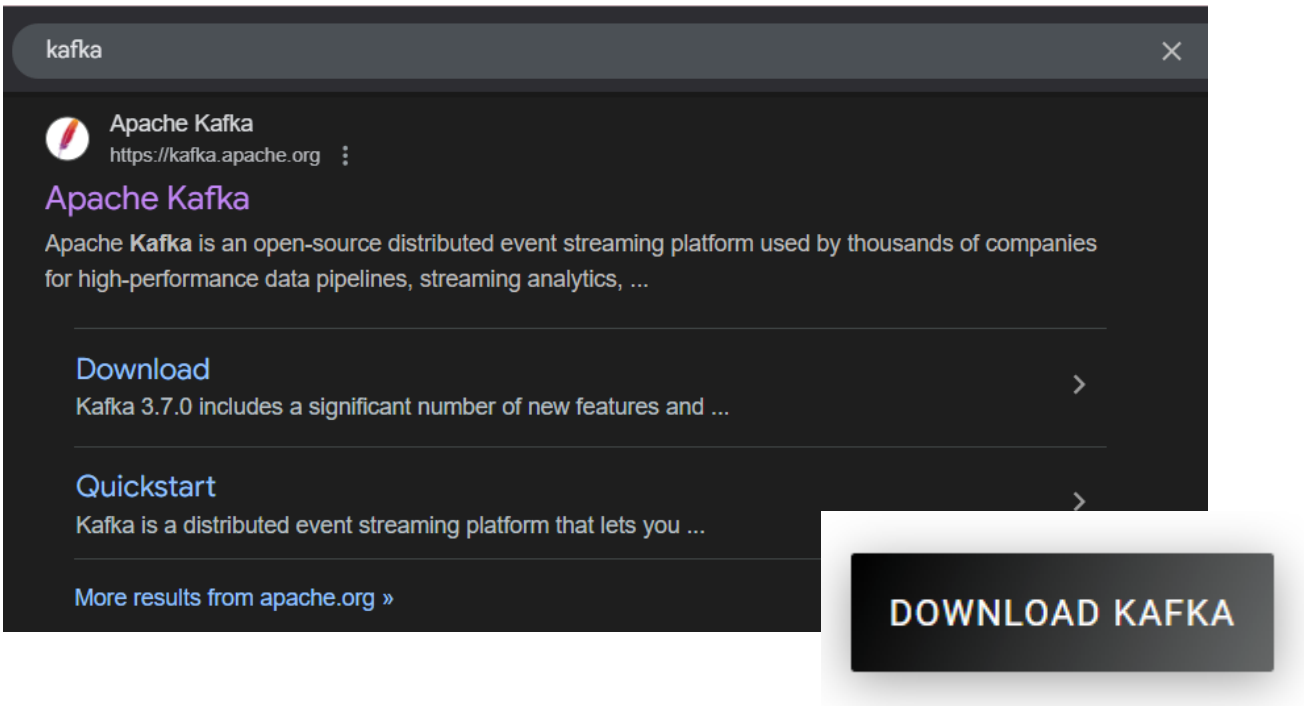


Navigating down below using the arrow keys, additional text was added to the bottom of the file. The JAVA_HOME environment was set to the directory of the JVM. This was saved and exited.

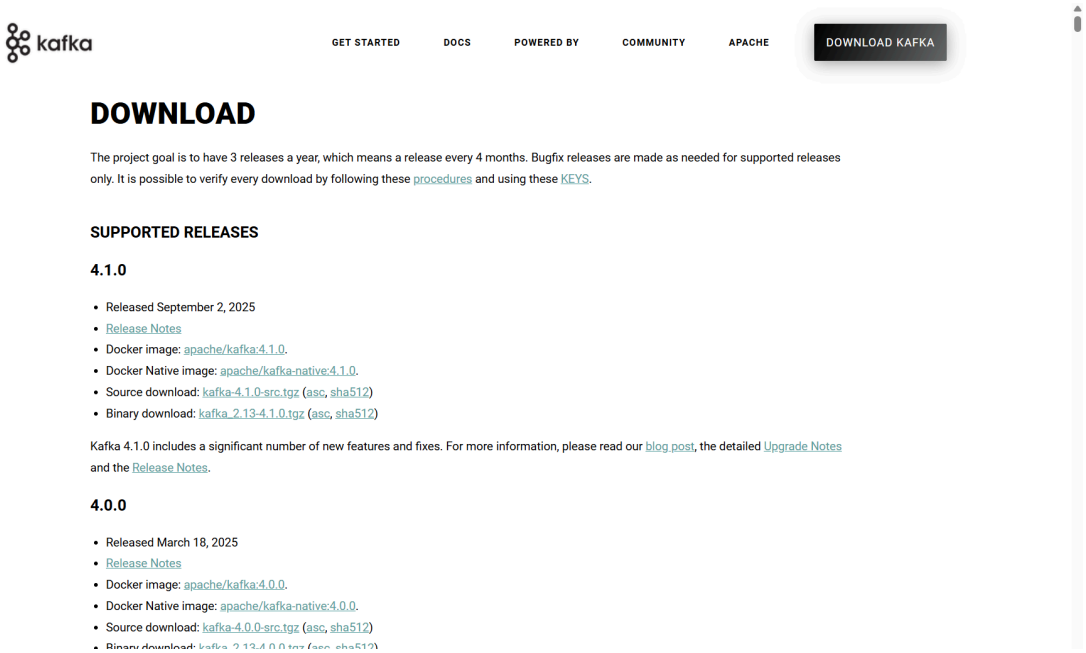


The changes in the file were run after running this code. '*Source*' executes the saved shell script in the shell environment (GeeksforGeeks, 2025)

Apache Kafka Installation



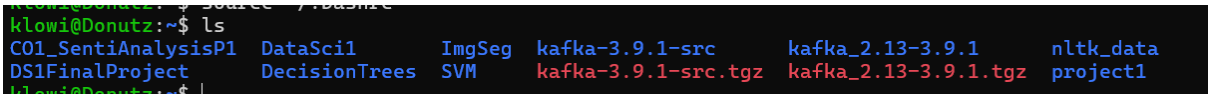
To start with installing Kafka, search for Apache Kafka in your desired browser, which will redirect you to the website. By the upper right of the home page, a *'Download Kafka'* button is available. Click this to be redirected to the Download page.



The Download Page contains all the recent releases of Kafka, alongside their information. During the current setup, the recent release was 3.9.1. There is a release every 4 months (Apache Kafka, 2025), so be sure to download the most recent one.



Look for the source download and get the link of your corresponding release. In the picture, the respective link and version was used. Use the “wget” command to download the Kafka binary sourced from the web (GeekforGeeks, 2024).

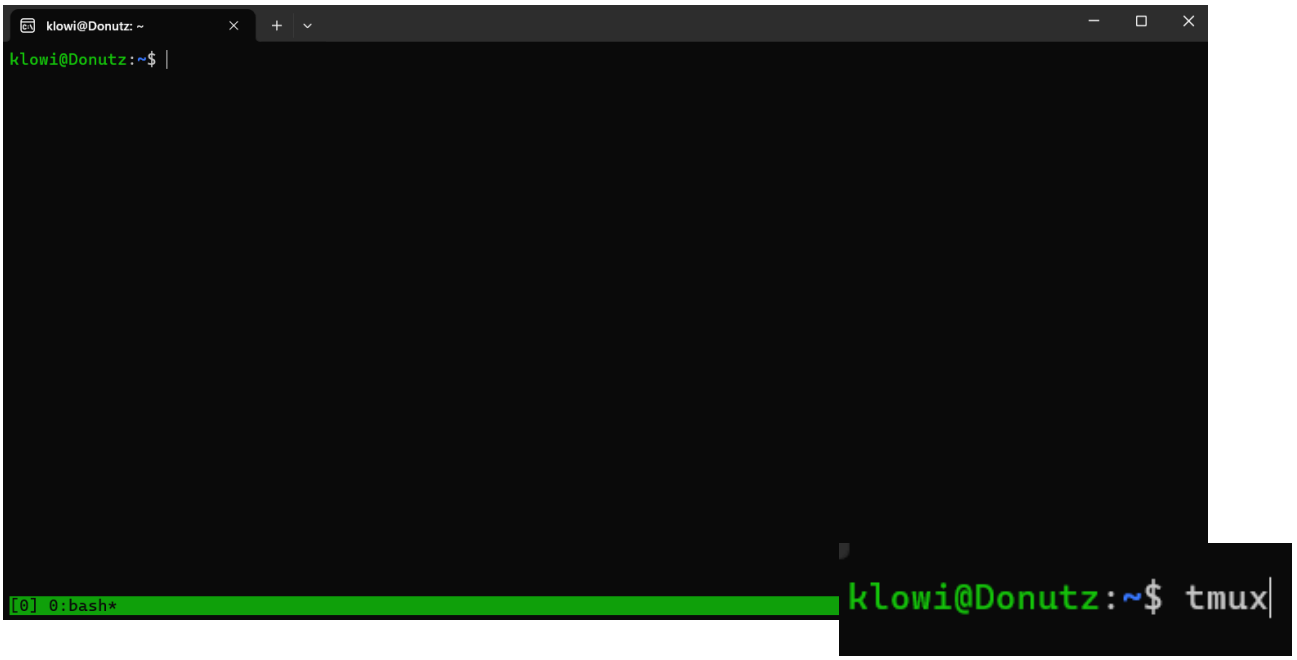


After downloading, it can easily be seen as a compressed archive *(kafka-3.9.1-src.tgz)*. Thus, it would require file extraction.

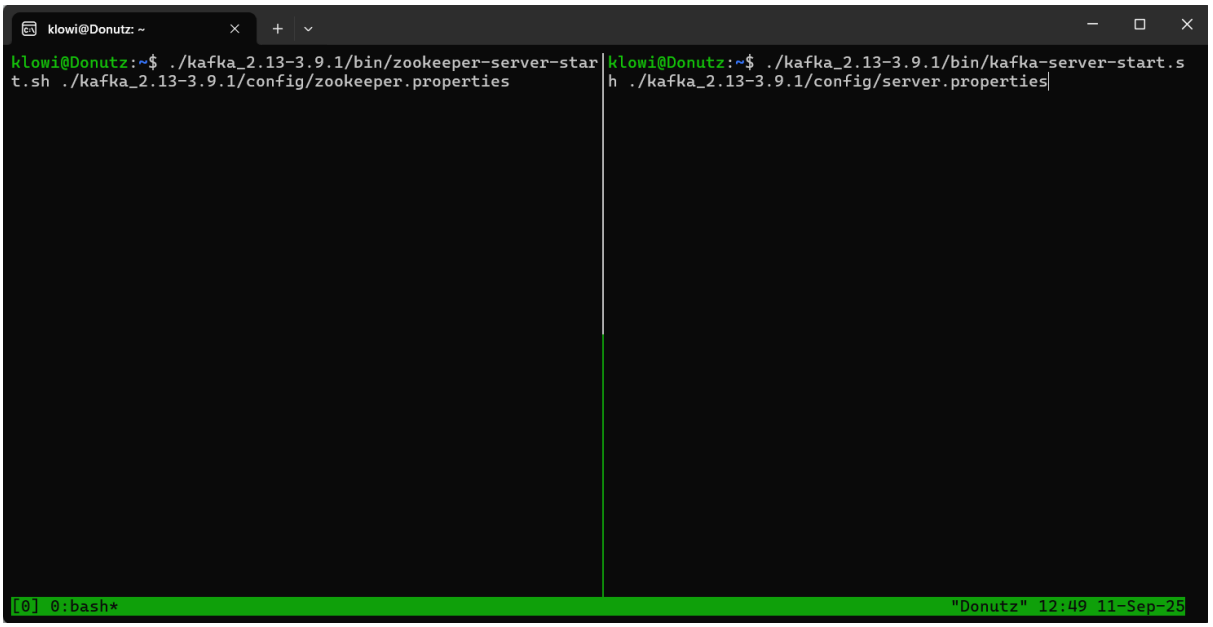


To extract, use the command “*tar -xvzf*” that calls the tape archive to extract the compressed file (GeekforGeeks, 2025). The command will be followed by the name of your compressed file. When you list directories again, you can now see the a kafka folder in blue text *(kafka-3.9.1-src.tgz)*.

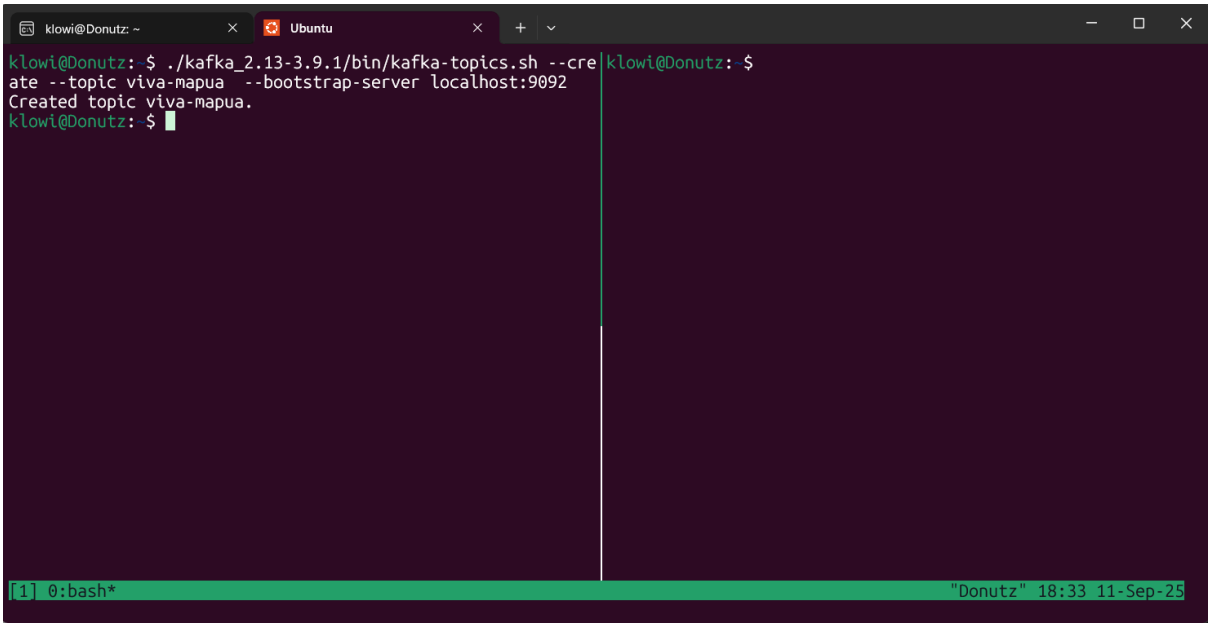
Starting Zookeeper and Kafka Servers



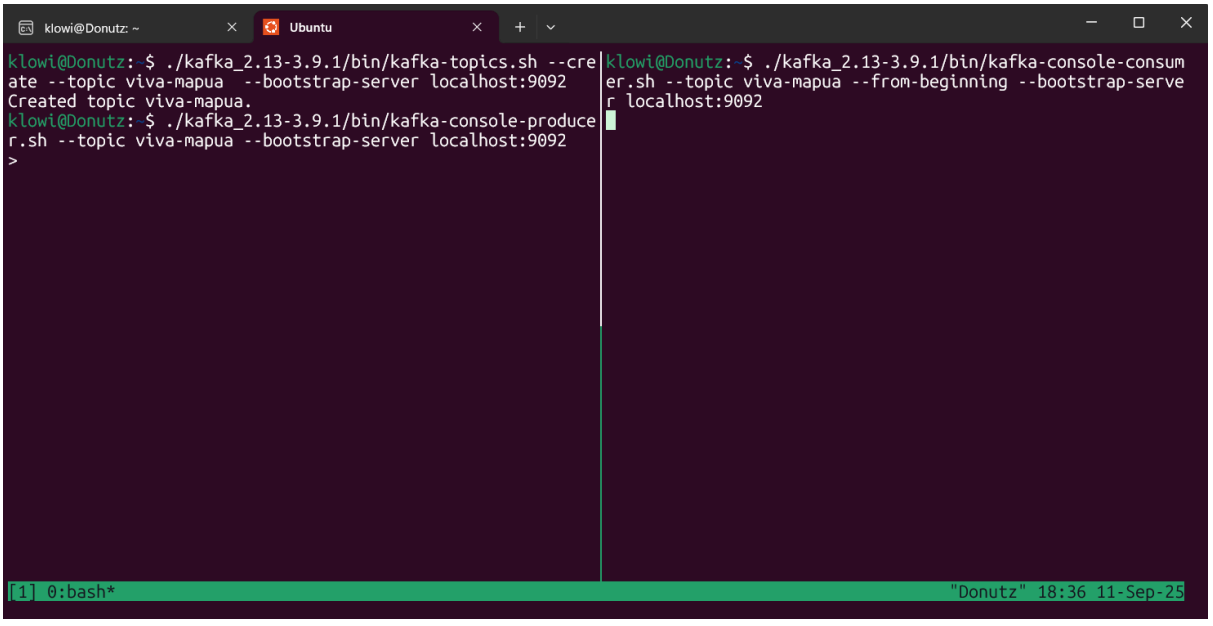
By now, you can turn on tmux in your WSL. Tmux is a terminal multiplexer that allows several terminal sessions in a single terminal (Gerardi, 2022).



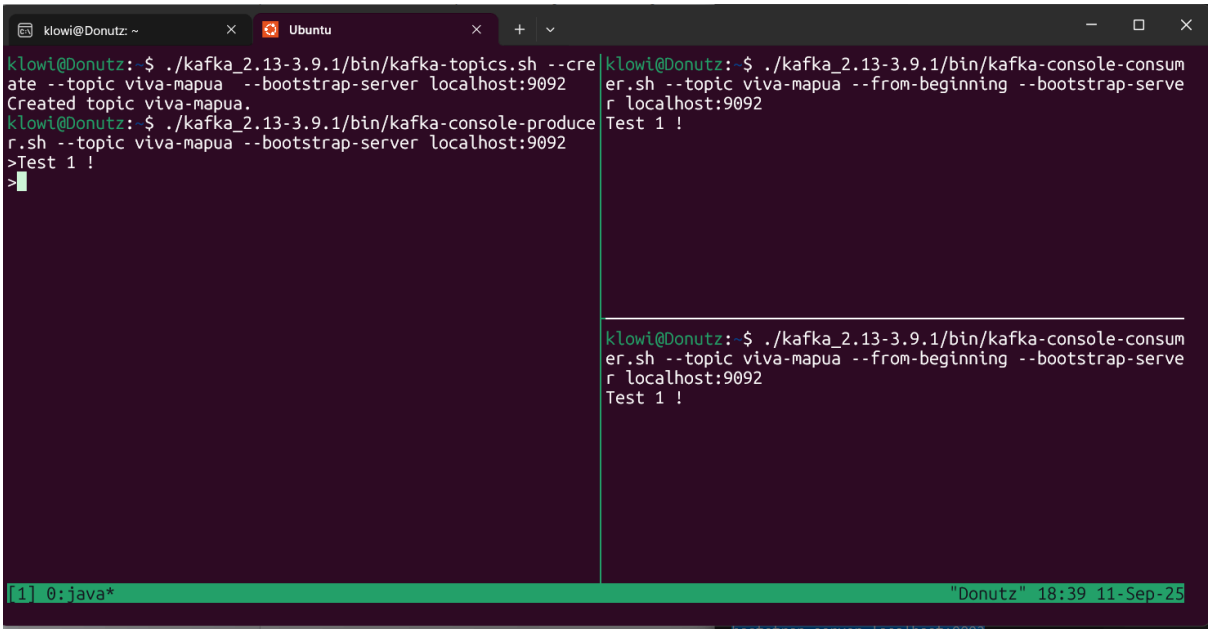
With the tmux on, you can run Zookeeper (left) and the Kafka (right) servers simultaneously. Press Enter on both terminal sessions to start.



In another pane, or terminal session (in this case, another terminal was created for organization and clear demonstration), create the topic that you want. In this assignment, the instructed topic was *"viva-mapua."*



Then, the following commands can be run to create the producer (left) and consumer (right). After this, producer messages that will be inputted can be easily seen in the consumer pane.



As seen above, multiple consumers with the same topic can be created. All messages in the prnoducer will be streamed and broadcast to the consumer end.

References

Apache Kafka. (2025, September). *Apache Kafka Releases*. Apache Kafka. Retrieved September 11, 2025

GeekforGeeks. (2024, April 23). *Wget Command in Linux/Unix*. GeeksforGeeks. Retrieved September 11, 2025

GeekforGeeks. (2025, April 11). *How to Compress Files in Linux | Tar Command*. GeeksforGeeks. Retrieved September 11, 2025

GeeksforGeeks. (2025, July 11). *Command in Linux with Examples*. GeeksforGeeks. Retrieved September 11, 2025

Gerardi, R. (2022, August 27). *A beginner's guide to tmux*. Red Hat. Retrieved September 11, 2025

Verma, J., Kurup, M., & guide, s. (2025, June 22). *What Is .bashrc in Linux? Configuration Explained*. DigitalOcean. Retrieved September 11, 2025